

CLAUDE.md — MTF Confluence Bot Project Rules

These rules apply to ALL actions Claude takes in this project — including every plugin, Claude Code skill (/statsmodels, /statistical-analysis, /time-series-forecaster, etc.), backtest script, data analysis tool, and any code generated or executed here.

MANDATORY GUARDRAILS

1. Data Source — yfinance only

- Use `yfinance` for ALL historical market data (prices, OHLCV, fundamentals).
- Never call the Alpaca API for historical or reference data.
- Never import `alpaca_trade_api`, `alpaca-py`, or any Alpaca SDK in scripts or skills.
- Exception: audit and diagnostic scripts may import bot modules (`strategy/`, `events/`, `indicators/`)

for read-only analysis — not for execution. No `TradingClient` or `StockHistoricalDataClient` may ever be instantiated outside of `execution/broker.py`.

2. No Execution Imports

- Never import or instantiate:

- `execution.broker`

- `execution.risk_manager`

- Any module that initializes an Alpaca `TradingClient` or `StockHistoricalDataClient`

- Backtest scripts must be self-contained. Audit/diagnostic scripts may import non-execution bot modules for inspection only.

3. Output to logs/ Only

- All script output, JSON results, charts, CSVs, and logs must be written to the `logs/` directory only.
- Do not write output files anywhere else in the project.
- **Exemption:** OS-level process lockfiles written to `/tmp/` (e.g. `fcntl`-based singleton locks) are exempt. These are not data output — they are system-level process coordination mechanisms that must reside in `/tmp/` by POSIX convention.

4. RTH Time Block

- All backtest scripts, data analysis tools, and skill-driven scripts must refuse to run during Regular Trading Hours on weekdays.
- Block window: **5:44 AM PDT to 1:00 PM PDT, Monday–Friday**
- Enforce this check at the top of every script before any work begins.
- Implementation:

```
from zoneinfo import ZoneInfo
from datetime import datetime
import sys
```

```
PDT = ZoneInfo("America/Los_Angeles")
_now = datetime.now(PDT)
if _now.weekday() < 5:
    _mins = _now.hour * 60 + _now.minute
    if (5 * 60 + 44) <= _mins < (13 * 60):
        print("BLOCKED: Cannot run during RTH hours (5:44 AM-1:00 PM PDT weekdays).")
        sys.exit(1)
```

APPROVAL REQUIREMENT

Before editing any code file, get explicit approval from the user.

Present the proposed change and wait for confirmation before writing anything.

This applies to ALL files — main.py, config.py, strategy/, execution/, events/, indicators/, utility scripts, and any new files created here.

No file is read-only from Claude's perspective — all files may be audited, diagnosed, and edited. The approval gate is the protection. When in doubt, show the diff and ask.

BOARD AUDIT PROTOCOL

This is a standing requirement — not optional, not deferred.

Proactive Audit Rule

The board must audit all files touched in a session AND all adjacent/dependent files before the session ends. A third-party agent should never surface bugs the board missed.

Specifically:

- Any build that modifies main.py triggers a full read of the modified functions plus all functions that call them and all functions they call.
- Any build that adds a new file triggers a read of all files that import it.
- Any build that changes data flow (signals, sizing, exits) triggers a cross-check of the full path: signal generation → scoring → sizing → entry → exit → P&L recording.

What "Audit" Means

- **TB (Thomas Peterffy, Brad Katsuyama, Kent Beck, Gene Kim, Wes McKinney, Yaron Minsky):**

Flag bugs at the assembly level. Exact failure conditions, edge cases, affected code paths.

Not "this might be a problem" — exact line numbers and reproducing conditions.

- **AB (Larry Harris, Ed Thorp, Arthur Levitt):**

Flag logic gaps that create risk exposure: slippage, P&L corruption, PDT rule violations, sizing errors, exit degradation.

- **BoD (Simons, Taleb, Peterffy, Kyle, Shaw):**

Flag architectural contradictions: dead code from conflicting builds, tail risk exposure,

invariant violations.

Audit Coverage Checklist (run at session end)

- [] All functions modified this session — read in full
- [] All callers of modified functions
- [] All new files — imports, side effects, output paths
- [] RTH block present in any new script that performs analysis or data fetching
- [] No raw requests calls to Yahoo Finance (use yfinance)
- [] No execution imports in analysis/backtest scripts
- [] GTC order handling — any orphan adoption path checked for stranded orders
- [] P&L recording paths — fill price source verified, no side ambiguity

BOARD INTELLIGENCE PROTOCOL

Permanent rule — applies to every board interaction in this project.

Intelligence Level

All 26 board members operate at **150 IQ** in their respective domains. This means:

- BoD outputs are graduate-research-level: rigorous, quantitative, no hedging on domain expertise.
- AB outputs lead with specific formulas, thresholds, and citations from published work.
- TB outputs flag bugs at the assembly level — not "this might be a problem" but exact failure conditions, edge cases, and tested mitigations.
- No filler. No "it depends." Every member gives a concrete directional answer.

Layman Fallback

If the user says "I don't understand that" or asks for clarification, Claude summarizes the board output in plain language without losing the directional conclusion.

Voting Rules

- **Strategy changes** (MIN_SCORE, stops, targets, exit logic): Full board vote — all 3 boards weigh in.
- **Feature-specific reviews**: Call the relevant domain board(s) only.
- Board must review actual data/logs before providing analysis — no fabricated facts.

Domain → Board Mapping (real verified members only)

Domain	Primary	Support
Signal construction, regime adaptation	Jim Simons (BoD)	David Shaw (BoD)
Position sizing, Kelly, kill switch	Ed Thorp (AB)	Nassim Taleb (BoD)
Market microstructure, order flow	Albert Kyle (BoD)	Larry Harris (AB)
Trade execution, entry/exit mechanics	Larry Harris (AB)	Peter Brandt (AB)
Compliance, PDT, regulatory	Arthur Levitt (AB)	—

Domain	Primary	Support
Macro regime, overnight holds	David Shaw (BoD)	Paul Tudor Jones (AB)
Cross-asset correlation, portfolio	Ray Dalio (AB)	Nassim Taleb (BoD)
ML/signals, overfitting	Marcos Lopez de Prado (AB)	Jim Simons (BoD)
Factor/momentum signals	Cliff Asness (AB)	Jegadeesh+Titman (AB)
ODTE / options / IV	Tom Sosnoff+Sinclair+Nathan (AB)	—
Trading psychology, exit discipline	Mark Douglas (AB)	—
Alpha validation	Igor Tulchinsky (AB)	—
Infrastructure, error handling	Thomas Peterffy (BoD)	Brad Katsuyama (TB)
QA, release readiness	Kent Beck (TB)	Gene Kim (TB)
Security, secrets, audit	Bruce Schneier (TB)	—
Data engineering, bar integrity	Wes McKinney (TB)	—
Async, low-latency	Yaron Minsky (TB)	—
Observability, alerting	Charity Majors (TB)	—
Quant architecture	Emanuel Derman (TB)	—

PROJECT CONTEXT

- **Bot type:** Alpaca Markets paper trading bot (MTF confluence scoring)
- **Account:** Paper account (~\$2.5K), PDT-constrained (max 3 day trades / 5-day window)
- **Scoring:** 12-point confluence system (live), 16-point system (log only, validation)
- **Profile active:** paper (INTRADAY_STOP_ATR_MULT=1.25, TARGET=2.5x, MIN_SCORE=9/12)
- **MRI active:** MacroRiskIndex — 8-component cross-asset score, 0–100, 5 levels
- **Plugins installed:**

- ~/.claude/plugins/claude-scientific-skills/ (K-Dense: statsmodels, statistical-analysis, etc.)

- ~/.claude/plugins/claude-code-plugins-plus-skills/ (Jeremy Longshore: time-series, regression, etc.)

- **Linked skills** (~/.claude/skills/): statsmodels, statistical-analysis, exploratory-data-analysis, matplotlib, polars

PENDING / DEFERRED

- C-2: paper=True hardcoded in execution/broker.py:21 — change to False only when going live
- P5 Bug Queue: 12 pre-existing bugs identified Apr 7 — see HANDOFF_2026-04-06.md for full list.

P5-C1 (FVG MultiIndex) and P5-C2 (run_movers ImportError) are highest priority.

P5-H3 (orphan GTC stranding) is live exposure with current open positions.